

ORIGINAL RESEARCH

How should covariates be handled in randomized trials? Empirical evidence from 50 trials and recommendations for practice

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Abstract

Background and Objectives: Covariate adjustment can improve precision and power in randomized clinical trials and is recommended by major regulatory agencies. However, there is limited empirical evidence on how different adjustment strategies perform across diverse real-world trials, leaving uncertainty about which methods and covariates should be prespecified in statistical analysis plans. We aim to address this gap and provide practical recommendations.

Methods: We conducted a large-scale empirical study using individual-level data from 50 publicly available randomized trials (29,094 participants; 574 treatment-outcome comparisons). We compared commonly used covariate-adjusted estimators, including analysis of covariance, inverse-probability weighting, g-computation, and machine learning-based approaches, combined with 3 covariate selection strategies. Performance was evaluated using precision gains, changes in point estimates, computational reliability, and the probability that covariate adjustment altered statistical significance relative to an unadjusted analysis.

Results: Covariate adjustment improved precision in most settings, with a median variance reduction of 13.3% for continuous outcomes and 4.6% for binary outcomes. Adjustment was substantially more likely to produce a gain in statistical significance than a loss. Parsimonious regression approaches using a small prespecified set of prognostic covariates performed as well as or better than more complex methods, particularly in small- to medium-sized samples. Machine learning-based estimators did not provide additional precision and were more prone to computational failure for binary outcomes.

Conclusion: Across a wide range of randomized trials, parsimonious covariate adjustment provided consistent efficiency gains without introducing systematic bias. These findings support routine covariate adjustment in primary trial analyses and provide practical guidance for writing statistical analysis plans, selecting covariates, and planning sample size. All curated datasets and analysis code are openly released as a reproducible resource to support future clinical research. © 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Keywords: Randomized clinical trials; Covariate adjustment; Statistical analysis plan; Precision; Prognostic variables; Causal inference

1. Introduction

In randomized clinical trials (RCTs), incorporating baseline covariates into the primary analysis (commonly referred to as covariate adjustment) has long been recognized as a principled way to improve precision and power without compromising the validity of treatment effect estimation. This view is supported by methodological research [1–4], regulatory guidance from the U.S. Food and Drug Administration [5] and the European Medicines Agency [6], and recent recommendations on estimands and statistical analysis plans [7,8]. Despite this

broad consensus, several practical questions remain unresolved: Which covariates should be included? How complex should the adjustment model be? And to what extent do different adjustment strategies change the conclusion of trials?

These questions arise routinely during trial design and the development of statistical analysis plans. The choice of adjustment model and covariates has direct implications for efficiency, robustness, and reproducibility. More flexible approaches, such as regression models with interaction terms [9] or machine learning-based estimators [10], can capture nonlinear relationships and treatment effect heterogeneity and may therefore improve precision. However, simpler models with a small prespecified set of covariates can offer greater stability and avoid overfitting in finite samples [11]. Clear empirical evidence comparing these approaches in real trials is therefore essential for informing best practice.

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Key findings

- Covariate adjustment improved precision in most randomized trials.
- Precision gains were greater for continuous than for binary outcomes.
- Simple regression often matched or outperformed complex estimators.
- Prespecified covariate adjustment should be considered routinely.
- A few prognostic covariates may be sufficient for primary analyses.

Most existing evidence on covariate adjustment is derived from asymptotic theory, simulation studies, or individual case studies. Although these approaches have generated important insights, they do not fully reflect the heterogeneity of sample sizes, covariate structures, and outcome-covariate relationships encountered in practice. Simulation results [12–14] depend on assumed data-generating mechanisms, and isolated case studies [15,16] are limited in scope and generalizability. As a result, trialists and applied statisticians still lack practice-oriented guidance on how covariate adjustment performs across diverse real-world settings and how it should be prespecified for primary analyses. To address this gap, we conducted a large-scale empirical evaluation of commonly used covariate adjustment strategies and provided practical recommendations for their use in RCTs.

2. Methods

2.1. Data curation

We searched 4 public data repositories, Dryad (<https://datadryad.org>), Harvard Dataverse (<https://dataverse.harvard.edu>), Zenodo (<https://zenodo.org>), and the R CRAN repository (<https://cran.r-project.org>) with keywords “randomized clinical trials,” “randomized trials,” and “randomized controlled trials”. Our objective was to identify 50 RCTs with accessible individual-level data, well-defined data structures, and accompanying data dictionaries. We excluded studies that did not focus on human health outcomes, were not individually randomized, had an unclear data dictionary, lacked an associated peer-reviewed publication, or were completed before 1990. This search was designed to assemble a broad and reusable benchmark dataset rather than to provide an exhaustive systematic review of all RCTs. As a retrospective empirical study based on publicly available trial datasets, this work

did not have a prespecified protocol, and no finalized reporting guideline was identified for studies of this type.

For eligible RCTs, we implemented a unified data-processing pipeline. Primary and secondary outcomes were defined according to the trial registry when available, or otherwise based on the primary publication. Treatment assignment variables were harmonized across studies. For each trial, we constructed a candidate pool of baseline covariates using variables prespecified in the primary statistical analysis, when available, and variables reported in the baseline characteristics table of the primary publication. The covariate adjustment strategies were then applied to this common candidate pool to enable consistent comparisons of adjustment methods across trials. Outcome variables with more than 40% missingness were excluded. Observations with missing values in the remaining outcome variables were omitted from the analysis. Missing baseline covariates were imputed using the mean for continuous variables and the mode for categorical variables.

2.2. Statistical methods

We compared covariate-adjusted estimators of the marginal average treatment effect for continuous outcomes and the marginal risk difference for binary outcomes. The estimators were grouped into 2 broad families.

The first family comprised commonly used regression-based approaches: analysis of covariance (ANCOVA) [17], analysis of heterogeneous covariance (ANHECOVA) [9,18], inverse-probability weighting (IPW) [19], and g-computation using logistic regression for binary outcomes (g-logistic) [20]. These methods are widely applied in primary analyses of clinical trials and are compatible with standard statistical analysis plans.

The second family consisted of estimators that incorporate machine learning–based outcome models. We included these approaches because they are increasingly advocated in the methodological literature for improving efficiency in RCTs. We considered targeted minimum loss estimation (TMLE) [10] and debiased machine learning (DML) [21]. For outcome prediction, both approaches used an ensemble of generalized linear models, penalized regression (GLMNET) [22], random forests [23], and Bayesian additive regression trees [24], with candidate algorithms selected based on predictive performance.

All covariate-adjusted estimators were compared with the unadjusted estimator, defined as the difference in mean outcomes between randomized groups. Technical details, software implementation, and algorithm selection are provided in [Supplementary Material B](#).

For each covariate-adjusted method, we examined 3 covariate selection strategies: *All*, *Top-3*, and *Baseline+*. The *All* strategy adjusted for all available baseline variables, maximizing the information used for prognostic adjustment but risking numerical instability when the covariate dimension is large relative to sample size. The *Top-3* strategy

Table 1. Statistical models and covariate selection strategies evaluated in this study

Estimators	
Unadjusted	Difference in mean outcomes between groups. No covariates involved.
ANCOVA	Classical regression adjusting for baseline covariates; estimates the treatment effect from a linear model including treatment and covariates.
ANHECOVA	Extension of ANCOVA that allows treatment-covariate interactions; implemented using g-computation for improved flexibility.
IPW	Outcome comparison after reweighting individuals by the estimated probability of receiving their observed treatment based on a logistic model.
g-logistic	G-computation estimator obtained from a logistic regression of the outcome on treatment and covariates.
DML	Cross-fitted outcome and treatment models are combined using an influence function–based estimator.
TMLE	Machine learning–based predictions are refined via a targeted update step to yield a g-computation estimator.
Covariate selection strategies	
All	Adjust for all available baseline covariates.
Top-3	Adjust for the 3 baseline covariates with the largest absolute Pearson correlations with the outcome, computed within each trial after pooling treatment groups.
Baseline+	Adjust for a prespecified set of commonly recommended variables, including the baseline outcome, stratification factors, age, sex, and weight.

ANCOVA, analysis of covariance; ANHECOVA, analysis of heterogeneous covariance; DML, debiased machine learning; IPW, inverse-probability weighting; TMLE, targeted minimum loss estimation.

adjusted only for the 3 covariates most strongly correlated with the outcome, representing a data-adaptive approach that prioritizes prognostic strength while reducing model complexity. The *Baseline+* strategy used a prespecified set of commonly recommended variables, including the baseline outcome, stratification factors, age, sex, and weight when available, providing a transparent adjustment set. [Table 1](#) summarizes all considered estimators and covariate selection strategies.

2.3. Performance metrics

We evaluated performance using measures that reflect the practical consequences of covariate adjustment for trial analysis. All metrics were computed for each treatment-outcome comparison.

2.3.1. Precision

Precision gain was quantified using the proportional variance reduction (PVR) relative to the unadjusted estimator. This measure represents the percentage reduction in the variance of the treatment effect estimates and can be interpreted as the corresponding reduction in the required sample size to achieve the same statistical power. Positive values therefore indicate increased efficiency from covariate adjustment.

2.3.2. Change in point estimates

To assess whether covariate adjustment systematically altered the estimated treatment effect, we calculated the difference between adjusted and unadjusted point estimates, scaled by the SE of the unadjusted estimator. Values centered around zero indicate that adjustment introduces no systematic shifts in the estimand.

2.3.3. Operating characteristics for trial conclusions

To summarize the real-world impact of covariate adjustment on statistical inference, we evaluated 3 quantities: the error rate, the covariate adjustment gain (CAG), and the covariate adjustment loss (CAL). The error rate was defined as the proportion of analyses in which the software failed to return a result. CAG is the probability that a covariate-adjusted analysis yields a statistically significant treatment effect (two-sided tests of zero effect at the 0.05 level) when the corresponding unadjusted analysis does not, whereas CAL is the probability that statistical significance is lost after adjustment. These quantities directly describe how often covariate adjustment would change the statistical conclusion of a trial.

3. Results

3.1. Data curation results

We identified 50 eligible RCTs, including 38 from Dryad, 2 from Harvard Dataverse, 2 from Zenodo, and 8 from the R CRAN repository. These trials comprised 29,094 participants and 574 treatment-outcome comparisons, of which 445 involved continuous outcomes and 129 involved binary outcomes.

[Figure 1](#) summarizes the characteristics of the curated dataset. The included trials span a wide range of therapeutic areas, and sample sizes were predominantly conducted within the past 15 years and have been widely cited in the literature, indicating that the dataset reflects contemporary and clinically relevant trial settings. Trial-level details are provided in [Supplementary Table 1](#).

3.2. Precision

[Figure 2](#) summarizes the PVR achieved by each covariate adjustment strategy. Across all methods and

sample sizes, covariate adjustment improved precision on average, with a median variance reduction of 13.3% for continuous outcomes and 4.6% for binary outcomes. These improvements translate directly into proportional reductions in the sample size required to achieve a desired statistical power.

For continuous outcomes adjusting for all covariates, ANCOVA yielded the largest precision gains and rarely produced losses. These benefits were especially pronounced in small samples, where complex models were vulnerable to overfitting. ANHECOVA and IPW performed similarly to ANCOVA in medium- to large-sized samples but occasionally lost precision in smaller trials, likely because the inclusion of treatment-covariate interaction terms reduces the effective degrees of freedom and increases variance. Machine learning-based estimators (TMLE and DML) approached the performance of ANCOVA and ANHECOVA in large samples but were generally less precise in small samples.

When adjustment was restricted to the 3 most prognostic covariates (Top 3 strategy), the instability observed under the all covariates strategy was reduced, and the performance of ANCOVA, ANHECOVA, and IPW became similar across sample sizes. In contrast, the relative performance of TMLE and DML changed little under this strategy, indicating that excluding weakly prognostic covariates had limited influence on the efficiency of these estimators.

The Baseline+ strategy produced smaller precision gains than adjusting for all covariates and slightly smaller variance reductions than the top 3, consistent with its more conservative prespecified design. Nevertheless, overall precision improvements remained consistently positive across sample sizes, and differences between estimators became minimal in medium to large trials.

For binary outcomes, the overall patterns were similar, but the magnitude of precision gains was smaller. This

likely reflects the more limited information content of binary outcomes compared with continuous measures. Despite this attenuation, ANCOVA combined with the Baseline+ covariate set still achieved a median variance reduction of 10.6%, demonstrating practically relevant efficiency gains even for binary end points. In addition, g-logistic regression performed comparably to ANCOVA across the curated datasets, suggesting that both approaches can provide substantial precision improvements in practice.

Supplementary Figure 2 presents parallel analyses restricted to the primary outcomes of each RCT and shows similar results.

3.3. Point estimates

Point estimate comparisons are summarized in Figure 3. Across all settings, the point estimate shifts were approximately symmetric around zero, suggesting no systematic bias was introduced by the covariate adjustment methods. Compared with adjusting for all covariates, more conservative approaches (Top 3 and Baseline+) yielded more stability in estimate shifts. In addition, the adjusted and unadjusted estimators exhibited closer point estimates for binary outcomes than for continuous outcomes. This pattern may reflect the bounded nature of binary outcomes, which leaves less room for point estimate shifts, and the weaker outcome variation explained by baseline covariates in our empirical datasets.

Differences in point estimates generally arise from multiple sources and may not necessarily indicate bias. Taking ANCOVA as an illustrative example, Wang et al [25] showed that ANCOVA approximately subtracts from the unadjusted estimate the component of the treatment effect driven by covariate imbalance and its prognostic strength. Consequently, when the adjusted covariates are well balanced or weakly prognostic, the discrepancy between the ANCOVA and unadjusted estimators tends to be small.

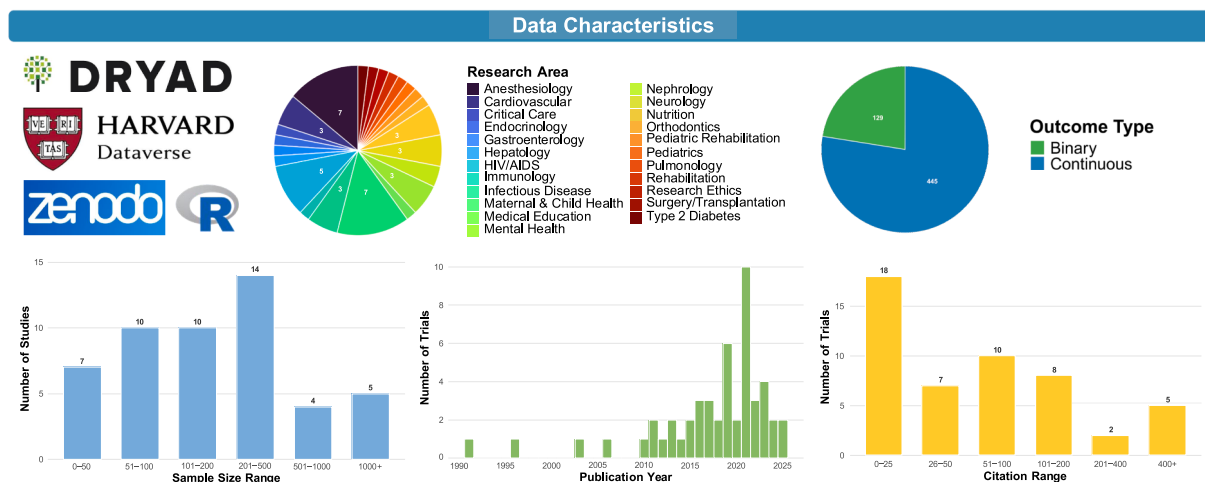


Figure 1. Characteristics of curated RCTs, including data sources, research areas, outcome types, sample sizes, publication years, and citation counts.

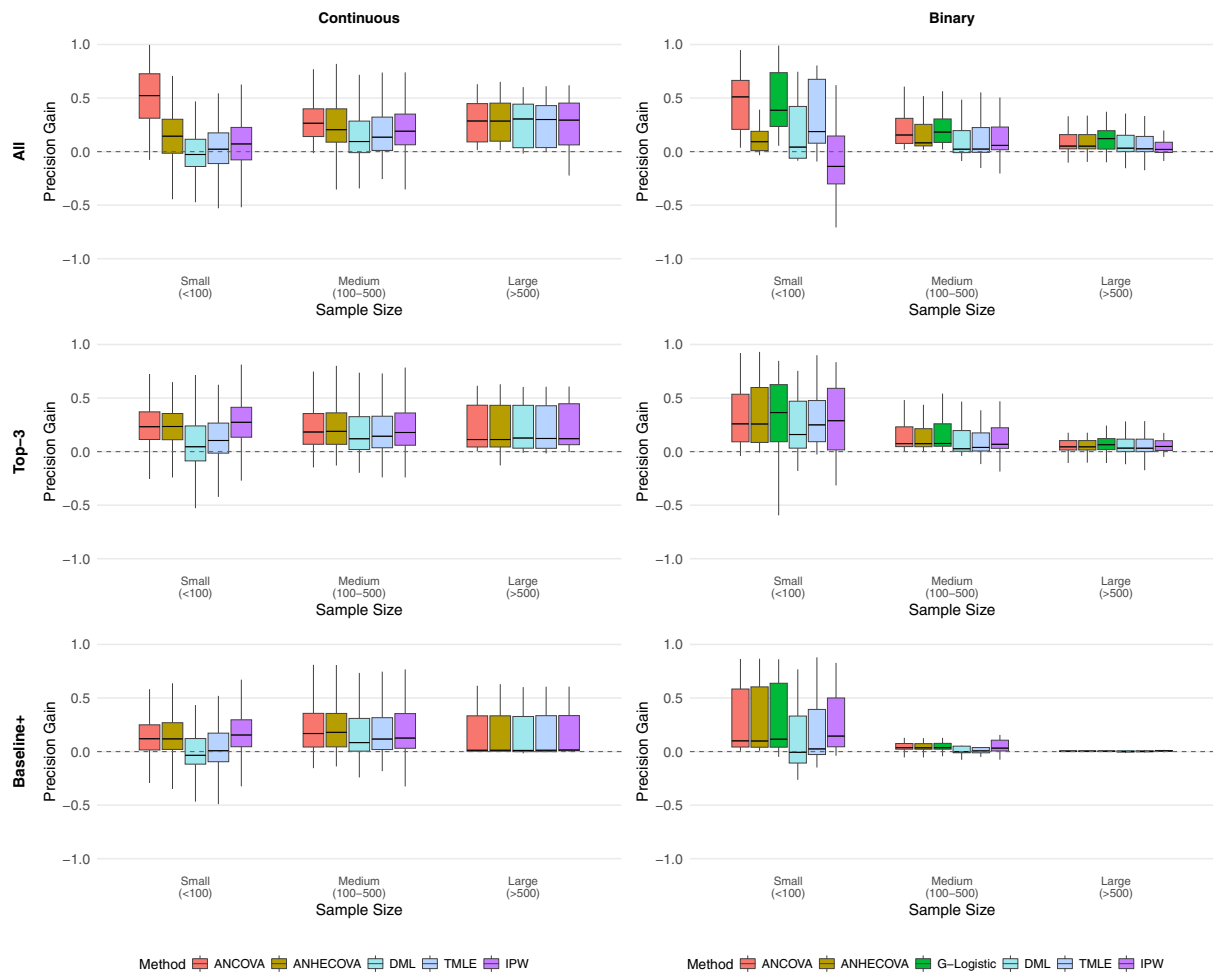


Figure 2. Box plots showing the empirical distribution of percentage precision gains across different sample size categories. In the 6 panels, columns separate outcome types, and rows represent covariate selection strategies, while colors denote the adjustment methods. The y-axis shows the proportional variance reduction (PVR) relative to the unadjusted analysis, with positive values indicating improved precision. The x-axis categorizes trials by sample size: small (0–100), medium (100–500), and large (>500). Other abbreviations used in the figure are defined in Table 1. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

For other estimators, the scaled difference may also depend on additional factors such as treatment effect heterogeneity and the allocation ratio.

3.4. Covariate adjustment gains and losses

As summarized in Table 2, covariate adjustment was generally advantageous across methods and outcome types: CAG consistently exceeded CAL, indicating that adjustment is more likely to create gains than to produce losses. For continuous outcomes, the largest benefits were observed for ANCOVA with all covariates (CAG = 17%, CAL = 4%), followed by ANHECOVA and IPW. For binary outcomes, gains were smaller overall, with the largest CAG observed for g-logistic (CAG = 11%, CAL = 2%). Machine learning–based estimators showed modest gains but low loss rates.

Patterns across covariate selection strategies reveal a trade-off between maximizing efficiency and controlling

instability. Adjusting for all covariates produced the highest CAG values but also slightly elevated CAL, consistent with increased sensitivity to small-sample variability. In contrast, conservative prespecified strategies such as Baseline+ yielded smaller but stable gains: across all methods, covariate adjustment under Baseline+ remained approximately 3 times more likely to yield a gain than a loss. Notably, some apparent losses may reflect the removal of spurious statistical significance arising from baseline imbalance, as discussed in Section 3.3.

3.5. Computation error rate

Table 3 summarizes the computational error rates for each method. Algorithm-specific nonconvergence warnings were partially reflected in the error rates and not separately reported. For continuous outcomes, all covariate adjustment approaches ran reliably and returned valid results for every treatment-outcome comparison. In contrast, several

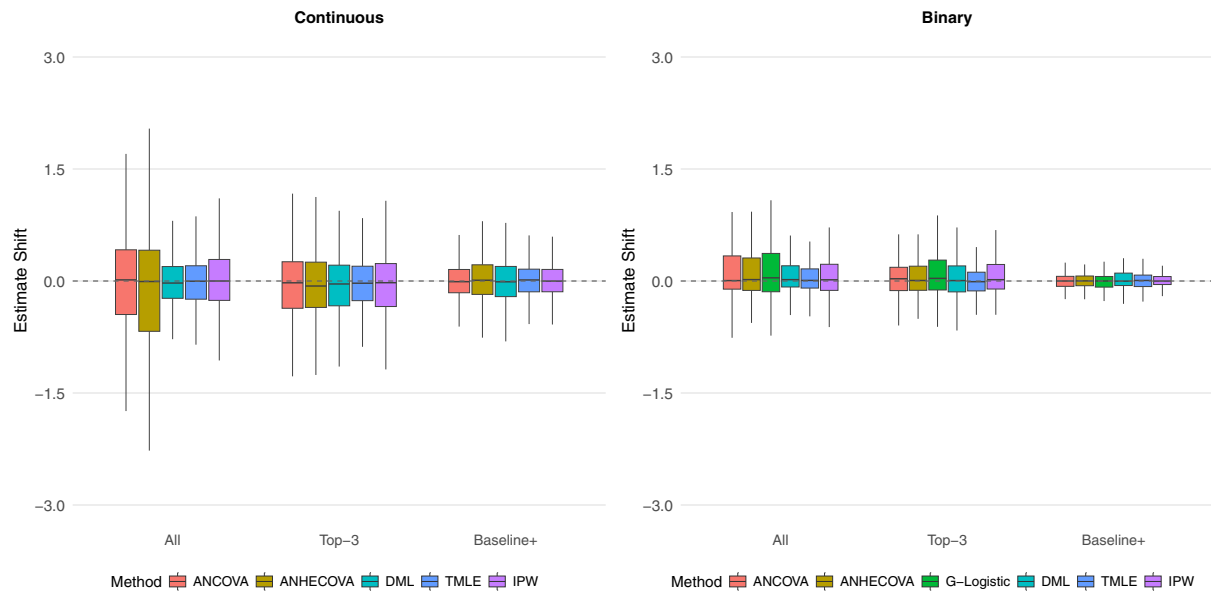


Figure 3. Box plots showing the empirical distribution of estimate shifts scaled by the SE of the unadjusted estimators across adjustment methods and covariate selection strategies. Column panels separate outcome types, and abbreviations used in the figure are defined in Table 1. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

methods exhibited nonnegligible error rates for binary outcomes.

Most computational failures were associated with sparse binary outcomes. When using logistic regression, rare outcome categories combined with a large number of covariates occasionally resulted in near-singular covariance matrices, yielding unstable SE estimates or convergence failures. For DML and TMLE, because they rely on cross-fitting, rare events may be absent within training folds, leading to model-fitting failure. TMLE was comparatively more stable because its widely used R implementation incorporates safeguards and fallback procedures that mitigate fitting failures. Overall, these

findings indicate that covariate adjustment for binary outcomes is more susceptible to computational instability than for continuous outcomes, particularly in the presence of rare events.

4. Discussion

As covariate adjustment becomes increasingly common in RCT analyses, investigators need practical guidance on choosing adjustment methods and covariate selection strategies. We addressed this need through a large-scale empirical benchmarking study using individual-level data from

Table 2. Covariate adjustment gain (CAG) and loss (CAL) across methods and covariate selection strategies

Outcome	Method	All		Top 3		Baseline+	
		CAG	CAL	CAG	CAL	CAG	CAL
Continuous	ANCOVA	17%	4%	6%	2%	6%	2%
	ANHECOVA	12%	7%	7%	2%	6%	2%
	IPW	6%	6%	7%	2%	6%	2%
	DML	7%	4%	6%	3%	6%	4%
	TMLE	5%	3%	5%	2%	5%	2%
Binary	ANCOVA	8%	1%	3%	2%	3%	1%
	ANHECOVA	4%	1%	3%	2%	3%	0%
	IPW	3%	0%	3%	0%	3%	0%
	G-logistic	11%	2%	7%	2%	4%	1%
	DML	4%	0%	3%	1%	2%	0%
	TMLE	1%	1%	2%	1%	2%	0%

ANCOVA, analysis of covariance; ANHECOVA, analysis of heterogeneous covariance; DML, debiased machine learning; IPW, inverse-probability weighting; TMLE, targeted minimum loss estimation.

Table 3. Error rates across methods and covariate selection strategies

Outcome	Method	All	Top 3	Baseline +
Continuous	ANCOVA	0%	0%	0%
	ANHECOVA	0%	0%	0%
	IPW	0%	0%	0%
	DML	0%	0%	0%
	TMLE	0%	0%	0%
Binary	ANCOVA	0%	0%	0%
	ANHECOVA	0%	0%	0%
	IPW	3%	2%	0%
	G-logistic	4%	1%	0%
	DML	12%	12%	8%
	TMLE	4%	4%	3%

50 completed RCTs. By analyzing observed trial data, our evaluation captured features difficult to represent in simulations and provides practice-oriented evidence to support recommendations for primary trial analyses.

Our findings showed that flexible, data-adaptive algorithms did not outperform classical parsimonious regression models for covariate adjustment in RCTs. A likely explanation is that many RCT datasets lack the structural complexity required for machine learning methods to deliver efficiency gains. Randomization reduces the need for extensive modeling of confounding, treatment effects are often well approximated by relatively simple outcome models, and typical trial sample sizes limit the stability of highly flexible estimators. Although these considerations are specific to randomized settings, similar patterns have been observed in observational studies, where simple regression approaches frequently match or exceed machine learning methods in terms of the bias-variance trade-off [26–28]. Concerns about the finite-sample reliability of data-adaptive estimators have also been noted in both methodological and applied work [29–32]. At the same time, machine learning methods remain valuable tools within the broader causal inference framework. They are particularly well-suited to settings with complex data structures or when the primary objective is to estimate heterogeneous

treatment effects [33]. The key challenge is therefore not to replace simple models with more complex ones, but to identify settings in which their advantages align with the available data.

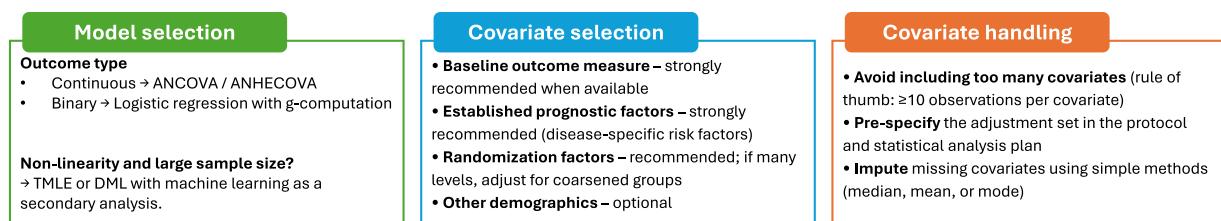
4.1. Practical recommendations

Our analyses provide clear insights for the prespecification and implementation of covariate adjustment in RCTs.

First, we recommend adjustment for prognostic baseline covariates to enhance statistical efficiency and power. These gains were consistently larger for continuous outcomes than for binary outcomes. Importantly, the probability that covariate adjustment led to a gain in statistical significance exceeded the probability of a loss, indicating a net benefit in practice.

Second, we recommend parsimonious regression methods, such as ANCOVA, for covariate adjustment in primary analyses. In our empirical evaluation, increased model flexibility did not translate into practically relevant precision gains. Differences between estimators diminished as sample size increased, while in smaller trials, simpler models consistently performed better because of their numerical stability and robustness. For variance estimation, we recommend using robust (sandwich) SEs that provide asymptotically valid inference even when the working outcome model is misspecified [5,34]. However, we note that sandwich variance estimators may be inaccurate in small samples, and their finite-sample performance can depend on the design and correction methods [35].

Third, we recommend prespecifying a small set of clinically established prognostic covariates. The Baseline+ strategy produced precision gains comparable to those of data-driven selection while avoiding the risks of post hoc variable selection and maintaining regulatory transparency. This approach is broadly applicable across therapeutic areas and can be tailored using subject-matter knowledge. For example, in diabetes trials, duration of diabetes and glycated hemoglobin are well-established prognostic factors and natural candidates for additional adjustment [36]. Although adjusting for all available covariates yielded the

**Figure 4.** Practical recommendations for covariate adjustment.

largest gains in some settings, it was more prone to computational instability in small samples.

Figure 4 summarizes these recommendations as a practical workflow for choosing adjustment models, selecting covariates, and handling covariates in statistical analysis plans. Across settings, the key principles are prespecification, parsimony, and prioritization of established prognostic variables.

4.2. Limitations

This study has several limitations. First, the curated dataset was assembled from a nonexhaustive search of publicly available RCTs. Although it spans multiple therapeutic areas and a wide range of sample sizes and designs, it may not fully represent the entire spectrum of modern clinical trials. In addition, many of the 574 treatment-outcome comparisons were derived from the same trials and therefore share baseline covariate information, inducing correlation among observations. However, analyses restricted to primary outcomes (Supplementary Fig 2) yielded consistent results. Accordingly, our findings should be interpreted as evidence of broad empirical patterns rather than predictions for any specific trial or disease setting.

Second, we evaluated a focused set of covariate adjustment methods to facilitate transparent comparisons. However, other approaches may perform better under particular conditions. For example, overlap weighting has been shown to improve stability relative to traditional IPW [37]. We additionally evaluated these types of methods and presented the results in the Supplementary Material. Hyperparameter tuning may enhance the performance of machine learning-based estimators [21]. In RCTs, however, such tuning is often constrained by prespecification requirements and the need for reproducible analysis plans.

Third, our analysis focused on marginal average treatment effects for continuous and binary outcomes. Although this estimand is central to most regulatory and confirmatory trials, alternative estimands, such as conditional average treatment effects, marginal risk ratios, odds ratios, or survival-based measures, may interact with covariate adjustment differently [38]. Extending empirical evaluation to these settings would provide a more comprehensive understanding of the operating characteristics of adjustment strategies.

Finally, our evaluation did not incorporate several common design and data features, including missing outcomes and covariate-adaptive randomization. Missing data are ubiquitous in clinical research and are typically addressed using principled methods combined with sensitivity analyses to assess robustness [39,40]. Covariate-adaptive randomization procedures, such as stratified randomization, are also widely used and can improve efficiency when paired with appropriate covariate adjustment [41–43].

Incorporating these elements into future empirical studies will further clarify how design features and adjustment strategies interact in practice.

CRedit authorship contribution statement

Yulin Shao: Software, Methodology, Data curation. **Liangbo Lyu:** Validation, Methodology. **Menggang Yu:** Writing — original draft, Validation. **Bingkai Wang:** Writing — original draft, Supervision, Conceptualization.

Declaration of competing interest

B.W. reports financial support was provided by National Institutes of Health. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112374>.

Data availability

I have put the link to data and code in the manuscript.

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